

Fused framework for glaucoma diagnosis using Optical Coherence Tomography (OCT) images

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ABSTRACT

Glaucoma is the second leading cause of vision loss induced by impairment of the optic nerve. Manual procedure for diagnosing glaucoma is associated with several challenges such as human error and time consumption, due to which an alternative technology assisted diagnosis methodology is preferable. The preponderance of research involves machine learning or deep learning for classification of glaucoma, and employs the commonly used imaging modalities, fundus imaging and Optical Coherence Tomography (OCT) for classification of an eye as Normal or Glaucomatous. A fused framework has been proposed for the same, which involves fusion of predicted classes from machine learning and deep learning techniques using Major Voting and Weighted Decision Fusion. This research work further explores the variability in predicted classes and identifies an optimum prediction threshold for comparison with existing techniques. Based on experimentation performed, a Precision of 0.95, Recall of 0.97 and F1-Score of 0.96 have been observed for the proposed Major Voting Fusion Framework, which outperformed k-Nearest Neighbours (0.93, 0.97, 0.95), Random Forest Classifier (0.94, 0.92, 0.93), Support Vector Machines (0.92, 0.94, 0.93) and 3D-Convolutional Neural Network (0.91, 0.91, 0.91) based deep learning approach. Gradient Weighted Class Activation Maps (Grad-CAMs) are also plotted to assist medical experts in diagnosis. The proposed framework could be beneficial for examining glaucomatous patients using OCT images. The higher F1 Score also indicates robustness of our approach on an uneven number of normal and abnormal samples in the dataset as well.

1. Introduction

Glaucoma is a retinal disease that damages the optic nerve due to the buildup of excess fluid in the anterior part of the eye. The excess of fluid produced in the eye leads to increased intraocular pressure, thereby causing irreversible blindness. It occurs in two forms: Primary Open-angle Glaucoma (POAG) and Angle-Closure Glaucoma (ACG). POAG is a common form of glaucoma caused when the eye fails to drain the excess fluid, leading to aggravation of pressure which damages the optic nerve. This type of glaucoma is painless and goes unnoticed as it shows no change in the vision at earlier stages. ACG results in the intraocular pressure escalate suddenly due to the iris and cornea being very close hence blocking the drainage angle for the fluid in the eye. The blocking of the drainage angle causes a rise in pressure in the eye. Its signs include blurry vision, severe eye pain, headache, etc. This type of glaucoma can cause blindness if left untreated at the initial stage. Thus, in both cases,

regular examination of an eye could play a significant role in initial diagnosis (Boyd, 2019a; Boyd, 2019b), and treatment at the initial stages will deter blindness.

Recent studies have explored the effect of diabetes as a risk factor for glaucoma. Observations conclude that 36% of patients with diabetes are likely to be affected by glaucoma (Zhao and Chen, 2017). High blood sugar levels caused due to diabetes damage the delicate vessels present in the retina responsible for the passage of fluid through the eye. Hence, the blockage of these vessels fails to drain the excess fluid, leading to aggravation of pressure which damages the optic nerve and causes open-angle glaucoma (Hersh, 2020).

Fig. 1 shows the vision of a healthy (normal) eye and a glaucomatous (abnormal) eye (Tek Vision Center, 2016).

Initial diagnosis performed by Ophthalmologists includes measuring eye pressure, inspecting the drainage angle of an eye, testing the peripheral (side) vision, measuring the thickness of the cornea, etc. (Boyd,

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2020). Further, treatments include medications to lower the eye pressure at initial stages and laser surgeries such as Trabeculoplasty for POAG and Iridotomy for ACG. In the case of Trabeculoplasty, a laser improves the functioning of the drainage angle. Whereas in the case of Iridotomy, a hole is created in the iris to help the fluid flow to the drainage angle (Boyd, 2020). Glaucoma is the second leading cause of vision loss across the world after cataracts. Unlike cataracts, the vision loss caused by glaucoma is commonly irreversible and hence is an issue of concern. Tham YC et al. predict that the number of people affected by glaucoma is going to increase from 75 million in 2020 to 110 million in 2040, with Asia and Africa being majorly affected as compared to other countries (Tham et al., 2014). Further, a study of the Indian population revealed that older people and the ones with a family history of glaucoma are at a higher risk of occurring the same (Kavitha et al., 2014). Fig. 2 shows the estimated increase of glaucoma from 2010 to 2040.

The aforementioned techniques are manual and time-intensive. Experts use imaging modalities like retinal fundus imaging and Optical Coherence Tomography for diagnosis and monitoring of glaucoma. Thus, the use of Computer-Aided Diagnosis (CAD) systems can be an influential factor in scaling down the manual and time-consuming processes of diagnosis performed by ophthalmologists. Fundus or OCT images are fed into the CAD system to predict the class of the input eye image (OCT) as normal or abnormal (glaucomatous). Furthermore, it can be used for large-scale initial screening and act as a second opinion for ophthalmologists. Fundus imaging, a two-dimensional imaging modality, provides fewer details as compared to OCT which is a three-dimensional imaging modality. In fundus imaging, reflected light helps acquire a two-dimensional representation of the three-dimensional retinal tissues projected onto the imaging plane. OCT is a non-invasive imaging modality for cross-sectional tissue imaging and is constructed by measuring the time delay of reflected echoes and the magnitude of backscattered light. It uses infrared light based on low-coherence interferometry, which utilizes temporal and spatial coherence properties of optical waves scattered from tissues in the body of an organism (Podoleanu, 2005). The acquisition of appropriate OCT requires the patient to look at the internal target of the machine with a minimum pupil diameter of 3 mm induced by dilation. OCT is an analog of ultrasound imaging that produces a cross-section of retinal tissues in microstructures (Bhende et al., 2018). Fig. 3 shows the OCT image of healthy (normal) eye and glaucomatous (abnormal) eye.

1.1. Related work

Grewal et al. in 2008 gave Artificial neural network (ANN) based classification of OCT images into 'abnormal' or 'normal'. The structural features, namely S_{max}/I_{max} , cup-area and Retinal nerve fibre layer (RNFL) parameters were manually extracted by domain experts and fed into ANN classifiers. Based on experimental validation, classification

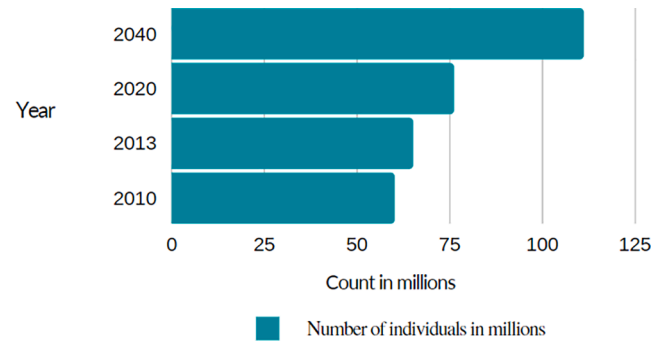


Fig. 2. Estimated increase of glaucoma worldwide.

using the extracted features and classifier improved the diagnosis with the specificity of 80% and sensitivity of 93.3% (Grewal et al., 2008).

Bizios D et al. in 2010 extracted RNFL parameters such as minimum/maximum RNFL thickness value, 10th/90th percentile and A-scan thickness measures for classification of OCT into 'glaucomatous' and 'normal' cases. Classification was further performed explicitly by the ANN and Support vector machine (SVM). On the basis of performance analysis of two classifiers, SVM was observed to slightly outperform Area under receiver operator characteristic (AUROC) of 0.98 (Bizios et al., 2010).

Schulze A et al. in 2011 suggested RNFL, Ganglion cell complex (GCC) and optic nerve head measures as a significant feature for diagnosis of glaucoma in OCT images. The thickness of each layer was computed for inferior and superior regions, followed by the average of the same. Each layer was then analysed individually, and the parameters of GCC were found to be slightly lower in diagnostic capability as compared to ONH parameters and RNFL thickness. Thus, the AUROC for RNFL parameters was found to be 0.828 and that for GCC to be 0.805 (Schulze et al., 2011).

Baskaran M et al. in 2012 used RNFL, Optic nerve head (ONH) parameters for classification of glaucoma using Linear discriminant analysis (LDA), Classification and regression tree (CART) classifiers. The AUROC for LDA was observed to be 0.98 and 0.99 for CART, and thus CART was found to outperform the LDA for diagnosis of glaucoma with RNFL and ONH features (Baskaran et al., 2012).

Barella KA et al. in 2013 investigated the performance of different machine learning classifiers, namely Naïve Bayes (NB), Bagging (BAG), SVM, Radial basis function (RBF), Multilayer perceptron (MLP), Random forest (RF), Ensemble selections (ENS), CART and Adaboost (ADA) for classification of glaucoma. The structural features such as RNFL thickness and ONH parameters were extracted for classifying OCT images as 'abnormal' and 'normal.' Further, based on the experiments performed, ADA was observed to show improved performance than



(a) Vision of a healthy eye



(b) Vision of glaucomatous eye

Fig. 1. Vision with eye.

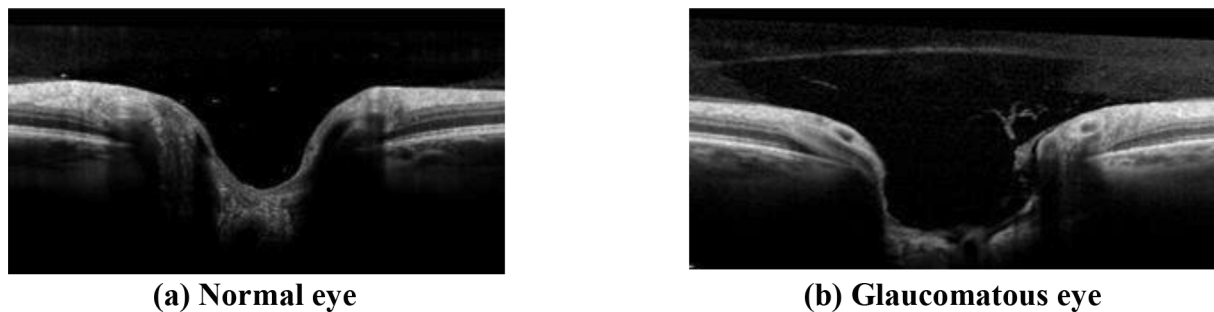


Fig. 3. OCT image.

other classifiers having AUROC of 0.83 (Barella et al., 2013).

Vidotti VG et al. in 2013 also presented an analysis of BAG, NB, SVM, RBF, MLP, RF, ENS, CART and ADA for classification of OCT into 'abnormal' and 'normal' cases. The features extracted for classifying included RNFL parameters, namely average thickness, temporal/superior/ nasal/inferior thickness and thickness at a clock hour of 1 to 12. In comparing the classifiers, BAG was found to outperform others with an AUROC of 0.81 (Vidotti et al., 2013).

Anantrasirichai N et al. in 2013 performed SVM-based classification of glaucoma in OCT images. The features, namely RNFL and GCC textures, were extracted, and Principal component analysis (PCA) was applied to convert features into linear correlated variables. Thereby, the OCT classification was done utilizing SVM, and a combined set of features were observed to improve the performance with an accuracy of 85.15% (Anantrasirichai et al., 2013).

Yoshida T et al. in 2014 gave a classification approach for glaucoma using an RF classifier in OCT images. RNFL and GCC parameters were extracted at different inferior, superior, nasal and temporal positions. The AUROC for the RF classifier was 98.5% at 95% confidence interval (Yoshida et al., 2014). Vlad S et al. in 2015 extracted structural features, namely RNFL thickness using OCT machines in inferior, superior, temporal and nasal regions for glaucoma diagnosis. Feed-forward neural network with weights and bias initialized using Nguyen-window was employed for classification, and training was done using the Levenberg-Marquard approach. The proposed approach for classification achieved 100% sensitivity and 94.3% specificity (Vlad et al., 2015).

Fu D et al. in 2016 initiated the denoising of OCT using Block matching and 3D filtering (BM3D) and extracted features such as retinal thickness, boundary continuity and boundary smoothness. These features were then used for classification of retinal abnormalities using a clustering-based approach. The specificity and sensitivity values were thus found to be 92% and 94% respectively (Fu et al., 2016).

Asaoka R et al. in 2017 employed RF for classifying retinal OCT into 'abnormal' and 'normal' cases. Structural features such as the thickness of GCC and RNFL at different inferior, superior, nasal, temporal positions were extracted and fed to the RF classifier for training to predict the outcome (normal/abnormal) on testing. The AUROC for the study performed was observed to be 93%, which was better than other state of the art methods such as bidirectional search, least absolute shrinkage selection operator regression and DeLong's method (Asaoka et al., 2017).

An et al. in 2019 used the fundus image, deviation map and thickness map of OCT image as input to train the VGG 19 network individually. Features extracted from each of these networks were then combined and fed into a random forest to classify normal and glaucoma cases. The AUC for this approach was found to be 0.96. (An et al., 2019).

Medeiros FA et al. in 2019 predicted the thickness of RNFL on the Spectral-Domain Optical Coherence Tomography (SD-OCT) based on the training provided using a deep learning approach. The value of mean RNFL thickness on SD-OCT was observed to be 82.5 ± 14.5 with AUROC of 0.944 (Medeiros et al., 2019).

Wang J et al. in 2019 gave a joint approach for segmentation and

classification of the retina from OCT based on RNFL thickness. A sequential multi-task convolutional neural network was employed to detect boundaries and segment retinal layers. The proposed BL-Net consisted of the B part for pixel-wise detection of retinal layers and the L part for segmentation of retinal layers. Interpolation was thereafter applied for refinement of boundaries and a bi-decision strategy was used for locating exact boundaries in the layered segmentation mask. On the basis of experimentation performed, the proposed approach gave an accuracy of 81.4% with AUROC of 0.864 (Wang et al., 2019).

Maetschke S et al. in 2019 performed direct classification of OCT images into 'normal' and 'abnormal' cases using a Three-dimensional convolutional neural network (3D-CNN) without extracting RNFL and GCC thickness. The proposed approach outperformed other state of the art machine learning classifiers with an AUROC of 0.94 (Maetschke et al., 2019).

Ran AR et al. in 2019 used a 3D deep learning approach for the classification of SD-OCT images based on residual networks. The comparison of demographic data was performed using Wilcoxon rank, and the AUROC for the proposed approach was found to be 0.96 with 89% sensitivity and 96% specificity. Further on the basis of heatmaps, areas with structural damage used for classification using deep networks were analyzed (Ran et al., 2019).

Russakoff et al. in 2020 proposed a network named gNet3D consisting of two FCN layers and three convolutional layers for classifying OCT cubes into normal and abnormal cases. The value of AUROC for gNet3D was found to be 0.88 (Russakoff et al., 2020).

Thompson et al. in 2020 also suggested a convolution approach for classifying glaucoma with OCT images. They observed that the segmentation-free deep learning based approach outperforms conventional RNFL thickness measurement for early screening of glaucoma. The values of AUROC were observed to be 0.96 with 95% specificity (Thompson et al., 2020).

George et al. in 2020 gave a guided attention-based 3D CNN method for glaucoma diagnosis and estimation of retinal structures. The AUC for this method was found to be 93.8%. (George et al., 2020).

In 2020, Wang et al. presented a semi-supervised joint structure and function multi-task learning for glaucoma screening using OCT images. Based on the performance of the presented approach, AUROC was observed to be 0.97 and 0.93 respectively for two different datasets (Wang et al., 2020).

Further, Table 1 summarises the approaches used for glaucoma diagnosis using OCT images.

From the state-of-the-art approaches, it can be observed that various features have been utilized now and again for classifying glaucoma with different classifiers. Hence it becomes an issue to identify the important features and classifiers utilized to diagnose glaucoma. Furthermore, machine learning techniques need better features as inputs to a classifier for improved training. Also, selecting important features remains a challenge in the case of classification. Therefore, convolution-based deep learning may solve these disadvantages by automatically extracting features to do self-learning. Presently, deep learning approaches have been used to classify glaucoma, as it may also not be able to learn

Table 1
Classification approaches used for glaucoma diagnosis.

Authors	Year	Features extracted	Classifier used	Datasets	Performance	
					Metric	Value
Grewal DS et al.	2008	RNFL	ANN	Local dataset	Specificity	80%
Bizios D et al.	2010	RNFL	SVM, ANN	Local dataset	Sensitivity	93.3%
Schulze A et al.	2011	RNFL, GCC, ONH	Statistical analysis	Local dataset	AUROC for SVM	0.828
Baskaran M et al.	2012	RNFL, ONH	LDA, CART	Local dataset	AUCfor CART	0.99
Barella KA et al.	2013	RNFL, ONH	RF, NB, RBF, MLP, ADA, ENS, BAG, SVM, CART	Local dataset	AUROC for ADA	0.839
Vidotti VG et al.	2013	RNFL	BAG, NB, SVM, RBF, MLP, RF, ENS, CART, ADA	Local dataset	AUROC	0.81
Anantrasirichai N et al.	2013	RNFL, GCC, textural features	SVM	Local dataset	Accuracy	85.15%
Yoshida T et al.	2014	RNFL, GCC	RF	Local dataset	AUROC	98.5%
Vlad S et al.	2015	RNFL	ANN	Local dataset	Sensitivity	100%
Fu D et al.	2016	Retinal thickness, boundary continuity, boundary smoothness	K means clustering	Local dataset	Specificity	94%
					Sensitivity	92%
					Specificity	92%
Asaoka R et al.	2017	RNFL, GCC	Random forest	Local dataset	AUROC	93%
An G et al.	2019	Fundus image, Thickness and deviation maps of OCT	VGG19, Random forest	Local dataset	AUC	0.96
Medeiros FA et al.	2019	RNFL	ResNet34	Local dataset	AUROC	0.944
Wang J et al. in 2019	2019	RNFL	BL-net	Local dataset	Accuracy	81.4%
Maetschke S et al.	2019	—————	3D CNN	Self-created public dataset	AUROC	0.864
Ran AR et al.	2019	—————	3D CNN	Local dataset	AUROC	0.94
Russakoff et al.	2020	—————	gNet3D	Private Dataset	Sensitivity	89%
					Specificity	96%
					AUROC	0.88
Atalie et al.	2020	—————	ResNet34	Local dataset	AUROC	0.96
George et al.	2020	—————	3D CNN	Private Dataset	AUC	93.8%
Wang et al.	2020	—————	Semi supervised joint structure and function multi-task learning	Local dataset	AUROC	Set 1 of dataset: 0.97, Set 2 of dataset: 0.93

all significant features due to slight differences amongst images. Thus, there is a need for an optimal approach of classification which considers all the features. Hence, this work presents a fused classification of machine learning and deep learning for optimal performance of classifiers.

1.2. Contribution

For the last few years, various attempts have been made in the sphere of CAD systems to realize the potential of deep learning and machine learning to create a state-of-the-art approach for glaucoma diagnosis. Furthermore, very few studies do an empirical analysis of predicted class fusion of either deep learning or machine learning based CAD systems. Thus, we propose a fused framework of deep learning and machine learning techniques. The fused framework presented in this paper combines the machine learning approaches: (1) SVM, (2) k-NN, and (3) RF with deep learning, based on convolutional neural networks (CNN) that perform self-learning using convolution layers. Two methods: (1) Major Voting and (2) Weighted Decision Fusion, are used to combine the classes predicted by all classifiers to conclude the final class of an OCT image as normal or abnormal. The fused decision ensures retrieval of discriminative information from multiple sets, resulting in avoidance of biased decisions. For a comprehensive understanding, a new term 'dispute' is introduced which measures the number of input samples for which dissimilar classes were predicted by the classifiers. Grad-CAM is also plotted to analyze the region of interest for the CNN model and to further aid clinicians in diagnosis. Experimental analysis was performed to find the ideal threshold to minimize disputes and predict the optimum final class for an input image.

Compared to past studies, we contribute the following: (1) Computer-Aided-Diagnosis for the classification of Glaucoma using both

machine learning and deep learning methods applied on OCT images. In contrast, previous methods have only made use of either machine learning or deep learning for the same. Moreover, there is a lack of literature exploring architectures on OCT imaging modality for glaucoma. (2) A fused framework approach to combine classes predicted by machine learning and deep learning. This proved to perform better than existing state-of-the-art methods as it is able to incorporate features extracted from both domains, potentially increasing the robustness of the proposed framework (3) A metric *dispute* for the evaluation of efficiency of the fusion of techniques to strengthen the foundation of our proposed approach. (4) Statistical methods are heavily used in the study to provide a set of 513 features from an OCT volume for feature selection which improves the performance of machine learning models.

The paper is further organised into Section 2 describing Methodology, Section 3 giving Experimental Setup and Dataset used followed by Section 4 Results, Section 5 Discussion and Conclusion in Section 6.

2. Methodology

The aim of the study is to classify input OCT images as normal or abnormal (POAG); this section describes the methodology formulated for it. The input image was at first pre-processed by wavelet filtering to eliminate the various types of noises existing in OCT images. Further, the Statistical features, Gray Level Run Length Matrix (GLRM) and Grey level co-occurrence matrix (GLCM), were extracted from the wavelet filtered image. Thereafter, significant features were selected using Gain ratio (GR), Information gain (IG) and Correlation statistical methods. Lastly, classifiers such as K nearest neighbour (k-NN), RF and SVM were used for classification. Moreover, the 3D-CNN was also utilized for performing classification, and weighted fusion of all classifiers for

accurate prediction was performed using majority voting. Fig. 4 presents the flow diagram of the proposed fusion methodology utilized for classification with detailed discussion in the following sections.

2.1. Input of OCT-volume

The dataset consisted of OCT Volumes. For the study, OCT modality was selected because: (1) OCT is a three-dimensional imaging modality and more detailed than fundus, hence it improves our fused framework approach as we are able to extract more features from OCT. (2) There is an abundance of literature which explores fundus imaging for Computer-Aided-Diagnosis of Glaucoma, hence this study aims at contributing to the less explored field of OCT volumes as an image modality for CAD of glaucoma. The advantages of OCT, which also contributed to its selection for the study are: (1) OCT imaging modality delivers high resolution images as it is based on light, not sound or radio frequency. (2) OCT is non-invasive and convenient for patients. (3) OCT is three-dimensional in nature and can produce images of the cross sectional view of tissue layers at micrometer resolution.

2.2. Preprocessing

Pre-processing is utilized to eliminate the noises residing in three-dimensional (3D) OCT imaging for improved diagnostics of glaucoma. Noise in the images may degrade its quality by suppressing the significant details that might lead to incorrect prediction of disease. Thus, a total of 8 wavelet filters in the combination of LLH, LHL, LHH, HLL, HLH, HHL, HHH and LLL were applied to the three channels of the image in order to remove the noise. Here, L stands for Low and H for High. This process resulted in the formation of 8 images for our original image. Wavelet filtering is found to remove the speckle noise from OCT images without losing significant edge details (Sahu et al., 2019). Further, normalization using division by 255.0 was applied to each image to make the images homogeneous for the deep learning convolutional neural network model (Richmond Alake (2020)).

2.3. Extraction of features (Van Griethuysen et al., 2017)

Features comprise important details to be utilized for classifying into

abnormal and normal cases. In this case, widely used features for classification include Statistical (18), GLCM (23) and GLRM (16) features, summing up to 57 for each image and adding up to 513 features per image when combined with different wavelet filters.

2.3.1. Statistical features

Statistical features are also used to find an image texture by measuring a histogram to get the likelihood of a pixel occurring in the image. The features here rely on the value of pixels instead of the interaction between neighbouring pixels. The histogram estimation in the first order is represented by $p(t) = \frac{N(p)}{P}$. Where t signifies the grey level in the image, P represents the number of pixels centred around the pixel expected in the neighbouring window, and $N(t)$ signifies the number of grey value pixels (b) lying in the range $0 \leq t \leq 1$. The Statistical features extracted here are Mean absolute deviation, Median, Minimum, Maximum, Mean, Robust mean absolute deviation, Root mean square, Range, Total energy, Skewness, Variance, Uniformity, 10 percentile, 90 percentile, Interquartile range, Entropy and Kurtosis.

2.3.2. Gray level Co-Occurrence matrix (GLCM)

GLCM is a technique to identify the texture of an image by measuring the frequency of pixels paired with values lying in the spatial resolution of an image. It is a matrix with size $M_g \times M_g$ signifying joint probability of second order in the image region having defined mask of $P(m,n|\delta,\theta)$. The (m,n) th place of the matrix signifies the occurrence of combined levels ' m ' and ' n ' occurring within two pixels separated by angle θ and distance δ . Further, $\delta=1$ results in two neighbours having 13 angles. The commonly extracted GLCM features include Inverse Variance, Cluster Prominence, Cluster Tendency, Cluster Shade, Joint Entropy, Difference Entropy, Difference Average, Difference Variance, Maximum Probability, Sum Entropy, Sum Squares, Autocorrelation, Joint Average, Contrast, Variance, Correlation and MCC.

2.3.3. Gray level run length matrix (GLRM)

GLRM represents a matrix that includes textural feature details to determine the object's texture. It may be characterized as a line of pixels having the same intensity values in a particular direction. The matrix is here calculated as $P(m,n|\theta)$ in varied directions θ , where each value is averaged computed over each direction. The (s,t) element of GLRM

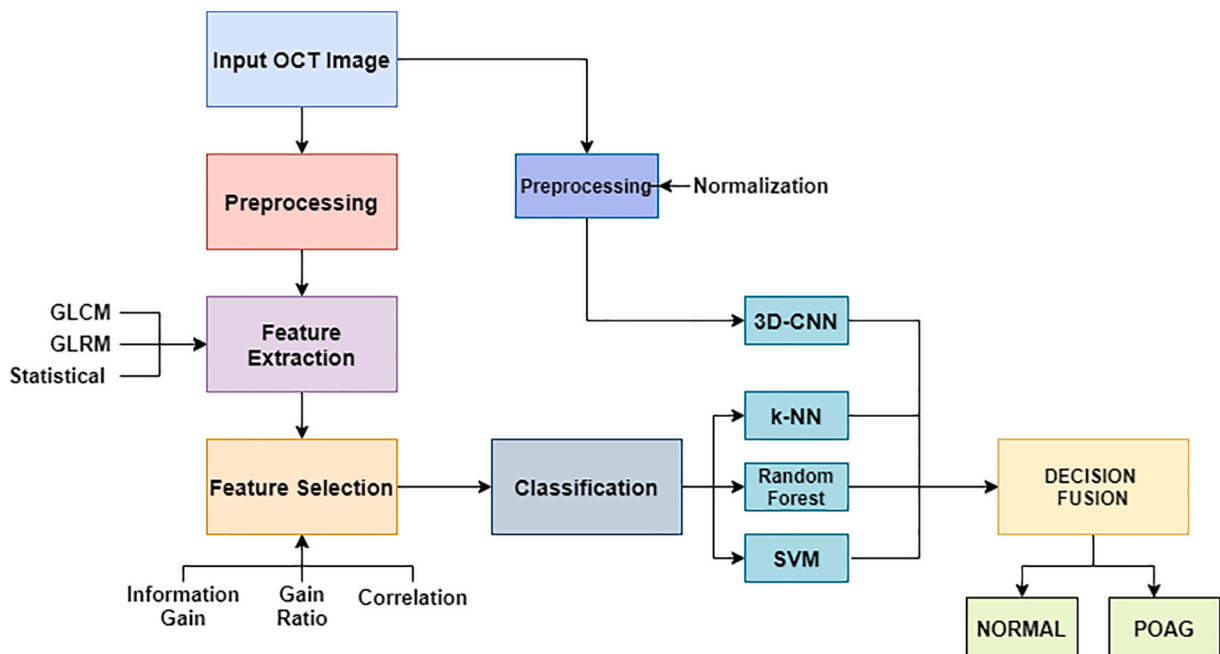


Fig. 4. Proposed Methodology.

represents the number of runs homogeneously for 'n' voxels having 'm' intensity in an image. The GLRM features extracted here for classification include Gray level non-uniformity, Gray level non-uniformity normalized, Long gray level run emphasis, Long run high gray level emphasis, Long run low gray level emphasis, Long run emphasis, High gray level emphasis, Run entropy, Run percentage, Run length non-uniformity normalized, Run variance, Short run low gray level emphasis, Short run high gray level emphasis, Short run emphasis, Run length non-uniformity, Run percentage and Gray level variance.

2.4. Selection of features (Holmes et al., 1994)

To further reduce the 513 features per image, Information Gain, Gain Ratio and Correlation based feature subset were used to infer the significant features for classification.

2.4.1. Information Gain (IG)

It is a method to rate feature information or entropy reduction (i.e. uncertain random features) for the class expected in the occurrence of adjacent data distributed with various features. Now for 'x' samples in set 'S' with the respective label 'y', the Eq. (1) is here used to calculate the measurement of IG for sample function.

$$IG(S, x_i) = H(S) - \sum_{v=values(x_i)} \frac{|S_{x_i=v}|}{|S|} H(S_{x_i=v}) \quad (1)$$

here, H(S) represents Shannon entropy.

2.4.2. Gain ratio (GR)

GR is employed to overcome the bias due to selection of attributes at every node in case of IG. It considers a sample 'x' having respective labels 'y' in a set 'S' for different values of 'I' features. It is given using Eq. (2).

$$Gain(S) = I(S) - H(S) \quad (2)$$

2.4.3. Correlation

It is the degree of resemblance amongst feature sets, while the significance of features in a feature set rises with correlation amongst features and class, thereby decreasing with increase in intercorrelation computed by Eq. (3).

$$r_{zc} = \frac{k\bar{r}_{zi}}{\sqrt{k + k(k-1)\bar{r}_{ii}}} \quad (3)$$

After applying the above techniques, a total of 140 different combinations of feature sets were obtained. Further, 513 features were reduced to the best performing 53 significant features as per scores of above selection approaches. Here 53 features included GLCM (19), GLRM (16) and statistical (18) on LLL wavelet filtered images.

2.5. Machine learning classification (Kononenko, 2001)

Classification using machine learning is prediction modelling based on learning with features extracted from images. The classification here is achieved by SVM, k-NN and RF.

2.5.1. Support vector Machine (SVM)

It is a supervised learning algorithm which plots feature inputs in a m-dimensional feature space for classification, where m is the number of features. The value of every feature corresponds to a value in a particular coordinate. The classification is performed by identifying a hyperplane to differentiate between two classes. Dataset for training is in the form $(s_1, t_1), \dots, (s_n, t_n)$, where the value of 't' is either 1 or -1 signifying the class pointed by 's'. The aim here is to find the optimal hyperplane that divides groups of s_i so that distance between hyperplanes separated by

't' and point 's' is maximized. The hyperplane is given as a set of points 's' satisfying $w \times s - b = 0$, where w is the vector normal to the hyperplane and b is the bias, often termed the Lagrange coefficient.

2.5.2. k-nearest neighbor (k-NN)

The nearest neighbour is a standard technique applied for statistical pattern analysis computed for the formation of a fixed sample space partitioning for s' observations categorized in the j^{th} class. The steps to identify s' are:

- Identify euclidean distance amongst s' and s in the feature set using $\sqrt{(s - s')^2}$.
- Select minimum 'k' distances.
- With these distances of k identify the most frequent class.
- The common class here is chosen as a class for new items.

2.5.3. Random forest (RF)

It is a classification algorithm that constructs tree classifiers at training time. The classifiers are formed by sampling a random vector from the input vector. RF training algorithms use general bootstrap or bagging techniques to help the trees in learning. Further, to train the set $S = s_1, \dots, s_n$ having labels $T = t_1, \dots, t_n$, bagging iteratively selects a sample at random by swapping training sets and fitting trees for samples. Random Forest is a common machine learning method that is heavily exploited as a data mining off-shelf technique.

2.6. Deep learning classification

The Convolutional Neural Network (CNN) approach operates directly on OCT Volume unlike machine learning which performs classification using manually selected features from the slices of OCT image modality. Following section explains the 3D-CNN used in detail.

2.6.1. Convolutional neural network (CNN)

The network consists of five 3D-convolutional layers with an Input Volume of 64x64x128, ReLU activation after each layer, followed by batch-normalization. Each Convolution layer involves a filter of sizes (7x7x7), (5x5x5) and (3x3x3) with stride size of (2x2x2) and (1x1x1), each layer having 32 numbers of filters. Global Average Pooling (GAP) is applied after the last convolution layer after which a dense layer with sigmoid activation is employed to predict class labels. Fig. 5 presents the block layered architecture of 3D CNN used to perform prediction. Training was performed using the best performing loss functions and optimizers such as Binary Cross entropy and NAdam having 0.001 rate of learning respectively. Further the classification as 'abnormal' (POAG) and 'normal' was done using logistic regression.

2.7. Decision fusion

Let *threshold* and *probability* be the variables, where *probability* is given by a model for a data point (single case) and *threshold* is a float value between 0.01 and 0.99. Further, the predicted class $\hat{Y}$ varies as:

$$Y = \{1 \text{ if } probability \geq threshold, 0 \text{ if } probability < threshold\} \quad (4)$$

To measure the performance of all feature sets, their F1-Score is calculated (using equation (9)) based on the predicted class given by equation (4) with *threshold* = 0.5 as a standard for all. The subset of features which gave the best performance were 53 features from the Wavelet-LLL category. *threshold* = 0.5 is used to initialize the approach and create a standard for all 140 combinations to decide the best subset. Equation (4) leads to the variation of the predicted class which in turn leads to a variation in F1-Score. The Wavelet-LLL feature set was then used for further machine learning, deep learning and their fusion

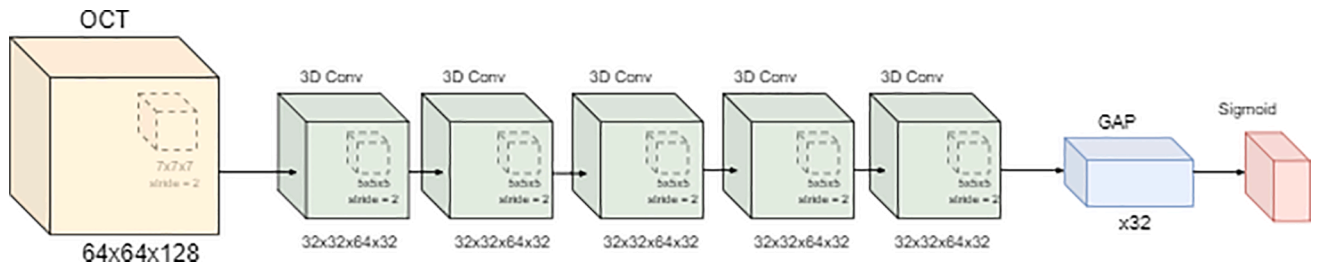


Fig. 5. 3D CNN Architecture.

according to the below-mentioned methods.

2.7.1. Weighted fusion

Every model was given a weight according to its F1-Score at a particular threshold. Here, weight is given as:

$$w_i = \frac{F1_i}{\sum_{i=0}^n F1_i} \quad (5)$$

here i represents the number of classifiers used and n represents the total number of classifiers used. In our case, it varies from 0 to 3 representing SVM, K-NN, RF and CNN. The predicted probability for Y_{prob} was then calculated as:

$$Y_{prob} = \sum_{i=0}^n w_i \times Y_i \quad (6)$$

where Y_i is the probability output from a model. The Fused predicted class was then calculated as per equation (4).

2.7.2. Major Voting

In Major Voting, output from a model was assigned equal weightage in the final decision. For major voting, predicted classes from each model were taken into fusion instead of probabilities. Class with highest count was declared as the fused output class $\hat{Y}$. For cases where the outputs conflicted such as two models predicting 1 and two models predicting 0, the model with the highest F1-Score was given the priority for final fused decision.

Further, the fusion of decisions based on two approaches mentioned in the above section was applied and F1-Score was plotted against the threshold (Fig. 6 shows the plots of F1-score vs. threshold for fusion of approaches).

From the plots of Fig. 6, it is evident that the fused prediction by Major Voting has outperformed every model in the approximate threshold range from 0.46 to 0.58.

2.7.3. Disputes

To further show that Information fusion has improved our approach, consider *disputes* to be the number of data points where the output class of any of the four models differed from each other. For example, [0,0,1,0], [0,1,0,1] etc. Plotting disputes before and after information fusion gives us the plots shown in Fig. 7.

In Fig. 7, IC after fusion stands for Incorrect Classification after fused decision. The vertical line shows the maximum difference between disputes before fusion and incorrect classifications after using fused decisions. It is observed that the implementation of fused decisions has decreased the disputes by 84%. Hence, it clearly shows that the use of fusion for output predictions from different models is a superior approach than singular models. Consequently, it is observed that the value of Fused F1-Score (by Major Voting) score at this maximum difference coincides with the maximum F1-Score achieved in the entire threshold range. Thus, the optimum value of threshold is chosen as 0.57 and final results are calculated using it. Further, Fig. 8 shows the confusion matrix and Area under receiver operating characteristics (AUROC) of 0.84 for fused decision.

3. Experiment setup and dataset

3.1. Experiment setup

The methodology presented for OCT classification into 'abnormal' (POAG) or 'normal' was executed in version 3.7.0 of Python along with version 2.1 of TensorFlow. Weka 3.8 was utilized for feature selection tasks (Hall et al., 2009). Also, the approach was executed on the Nvidia GeForce GTX TITAN XP GPU with Two Intel® Xeon(R) E5-2620 2.4 GHz CPUs.

3.2. Dataset used

The dataset used comprises 1110 OCT volumes with 847 glaucoma cases and 263 normal cases, each with dimensions $64 \times 64 \times 128$ voxels. The OCT scans in the dataset are acquired using the Cirrus SD-OCT

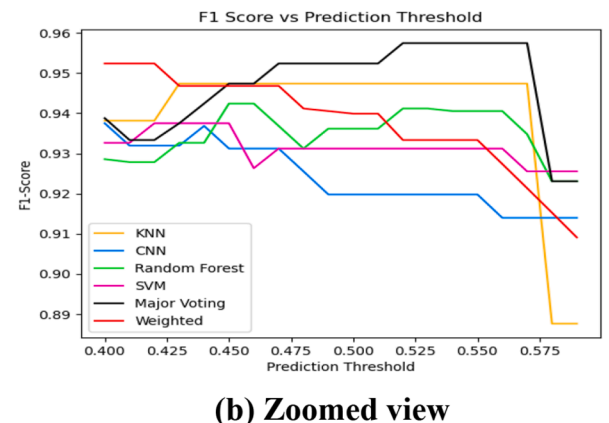
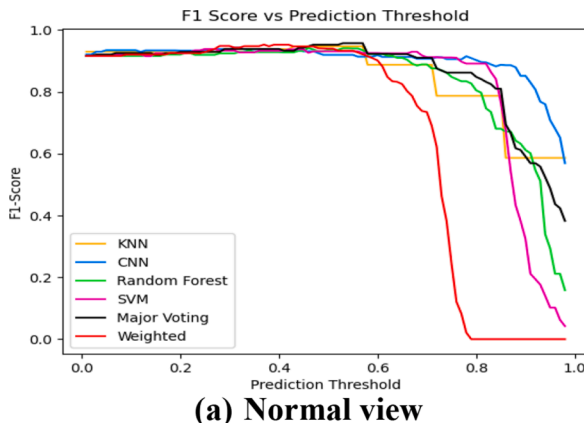


Fig. 6. Plots of F1-Score vs. threshold.

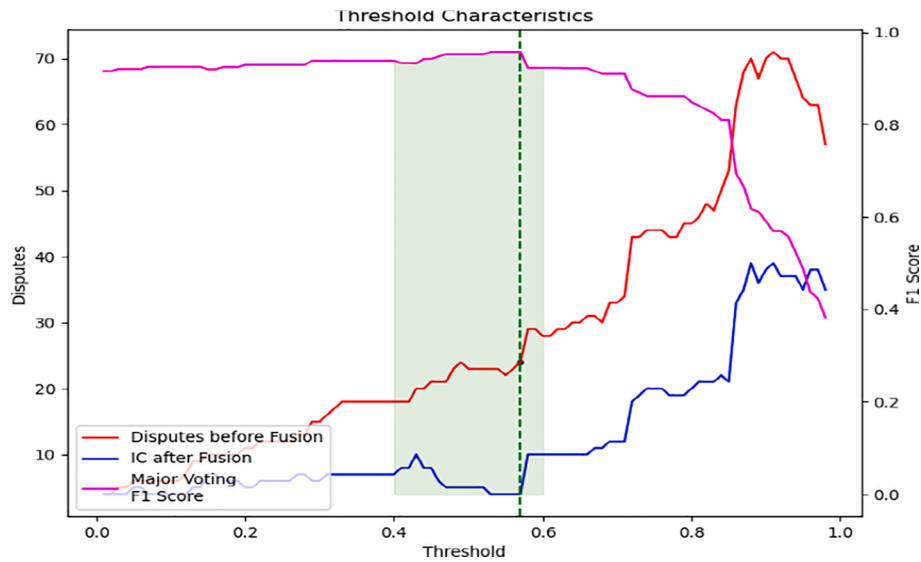


Fig. 7. Plots of Disputes.

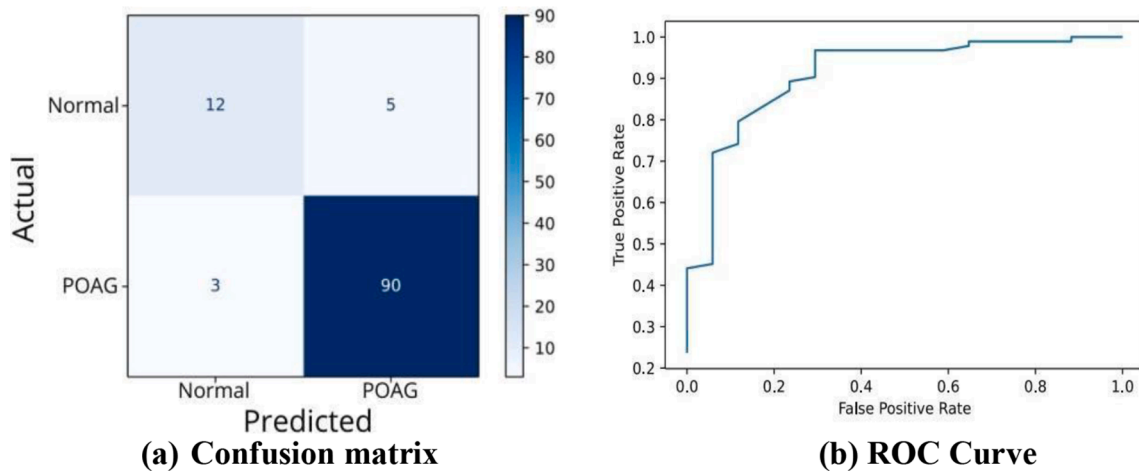


Fig. 8. Plots of performance for fused network.

scanner (Zeiss, Dublin, CA, USA). The dataset used for classification of glaucoma using OCT images in this study is publicly available at: <https://zenodo.org/record/1481223#.XYhpxCgzZPY> since 9th November 2018. The data set was split into 888 training samples, 110 test samples and 112 validation samples (80%, 10%, 10%). This was taken into consideration so that the eyes of the same individual were not split across folds.

4. Results

4.1. Performance parameters (Goutte and Gaussier, 2005)

The achievement of various classifiers was examined by parameters such as Precision, Recall and F1-Score. The evaluation parameters used are resultant of the values of True Positive (TP), False Positive (FP), True Negative (TN) and False Negative (FN) respectively derived by the confusion matrix shown in Table 2. Here,

TP are resulting cases where the model correctly predicts the positive class.

FP are resulting cases where the model incorrectly predicts the positive class.

TN are resulting cases where the model correctly predicts the negative class.

Table 2
Confusion matrix.

Classification outcome	Outcome of the ground truth (i.e. Abnormal/Normal)		
	Correct	Incorrect	Total of Row
Correct	TP	FP	TP + FP
Incorrect	FN	TN	FN + TN
Total of Column Values	TN + FN (Total cases having given outcome)	FP + TN (Total cases not having given outcome)	N = TP + FN + FP + TN (i.e. Total number of cases)

FN are resulting cases where the model incorrectly predicts the negative class.

Further Precision, Recall, and F1-Score can be computed as:

(i) **Precision:** It is the fractional measure of actual correct cases that are correctly identified as positive class and is calculated using Eq. (7).

$$precision = \frac{Tp}{Tp + Fp} \quad (7)$$

(ii) **Recall:** It is the fractional measure of actual correct cases that are correctly identified as it is and calculated by Eq. (8).

$$recall = \frac{Tp}{Tp + Fn} \quad (8).$$

(iii) **F1-Score:** It is calculated as the harmonic mean of Precision and Recall using equation (9).

$$F1 - Score = 2 \times \frac{precision \times recall}{precision + recall} \quad (9)$$

F1-Score is a robust metric which can be used for performance analysis of binary classification problems with a huge class imbalance in the dataset.

4.2. Parameters testing

The parameter selection for machine learning was done based on experimental analysis. As mentioned in Table 3, gamma for SVM was selected based on the given expression. For k-Nearest Neighbours it was observed that the model performed best with $k = 7$. For the Random Forest classifier, the number of trees was set to be 100 as it performed the best based on evaluation metrics. After careful inference of the trained deep learning CNN model on the validation data, the learning rate and epochs for training were finalized to minimize overfitting of data and maximise score on evaluation metrics. The parameters employed for testing and training of varied methods used in the study are given in Table 3.

Table 3 highlights the parameters used for SVM, k-NN and Random Forest. The data was split into training and testing sets to cross check the approach and all the results presented are on the testing data. The variation of F1 Score and AUROC (on training data) with respect to parameters of different ML models was studied and only the optimum parameter value was selected to avoid overfitting. The models used were never trained on the testing data, hence our study provides a firm evaluation of the fused framework proposed. The authors understand that the concern of overfitting is always present in the field of deep learning and the aim of study is to provide a robust model which can be applied on varied datasets.

4.3. Analysis of performance

The section discusses the performance of various classifiers which include k-NN, SVM, RF, 3D-CNN and proposed fused network on a full set and reduced set of features. Full set of features include 513 features and the reduced set include 56 features obtained after feature selection as discussed in 3.4. Thereafter, the analysis was carried out with k-fold cross validation having $k = 10$ which was considered by evaluation as shown in Table 4.

The graphical representation of Precision, Recall and F1-Score for reduced and full set of features on different classifiers is given in Fig. 9.

Further, Fig. 10 shows the comparative analysis of different classifiers on reduced sets of features.

Based on the experimental analysis the values of Precision, Recall and F1-Score for the k-NN classifier with full set of features are found to be 0.92, 0.88 and 0.90. Whereas, the values of the same for k-NN with reduced set of features are observed to be 0.93, 0.97 and 0.95 respectively. Likewise, Precision, Recall and F1-Score values for SVM

Table 3
Testing Parameters used for classification.

S. no.	Classifier	Dataset	Parameter Setting
1.	SVM	Original images	Radial Basis Function, gamma = 0.44 calculated using $1/\bar{A} \cdot (n_{features} \times X.var())$
2.	k-NN	Original images	$k = 7$ having Distance function as Euclidean
3.	Random Forest	Original images	With 100 numbers of trees.
4.	CNN	Original images	Optimizer NAdam with learning rate = 0.001 over 25 epochs

Table 4

Quantitative analysis of classification models.

S. No.	Architecture	Dataset	Performance metrics		
			Precision	Recall	F1-Score
1.	KNN (Full features)	Original images	0.92	0.88	0.90
2.	KNN (With reduced features)	Original images	0.93	0.97	0.95
3.	SVM (Full features)	Original images	0.92	0.92	0.92
4.	SVM (Reduced features)	Original images	0.92	0.94	0.93
5.	RF (Full features)	Original images	0.93	0.91	0.92
6.	RF (Reduced features)	Original images	0.94	0.92	0.93
7.	3D CNN	Original images	0.91	0.91	0.91
8.	Fused Network	Original images	0.95	0.97	0.96

classifiers with a full set of features are 0.92, 0.92 and 0.92; While, for a reduced set of features values of these metrics are 0.92, 0.94 and 0.93 respectively. Further the Precision, Recall and F1-Score for RF classifiers with full feature set are 0.93, 0.91 and 0.92; whereas, that for reduced feature set are 0.94, 0.92 and 0.93 respectively. Finally, the Precision, Recall and F1-Score for 3D CNN are 0.91, 0.91 and 0.91. While that for a fused network are 0.94, 0.97 and 0.96. Thus, the observations from the study are:

- Reduced set of features improves the performance of classifiers.
- KNN at 7 neighbours outperforms SVM, RF and 3D-CNN.
- Proposed fusion network improved the performance of classification.

Further, the trained CNN model was used to plot Gradient weighted Class Activation Maps (Grad-CAM) (Selvaraju et al., 2017) for input OCT slices along three different viewing axes. Figs. 11–13 show the three different views of the image slice and its Grad-CAM. Whereas, Fig. 14 shows the Grad-CAM views concatenated to show 3D voxels in axial view.

The red regions in the above figures show significant areas of interest for the CNN model where it may identify contours, texture etc., the intensity of interest decreases from red to orange to yellow and so on, where the blue highlights are non-significant regions. Grad-CAM can assist medical experts to identify the region of interests of the deep learning approach and hence aid in diagnosis.

5. Discussion

This study presented an approach based on the fusion of machine learning and 3D convolutional deep learning techniques in the domain of glaucoma classification using OCT Volumes. A few similar studies in this domain include an approach that used a fundus image, RNFL thickness map, RNFL deviation, GCC thickness and GCC deviation as input to train the VGG19 network individually. The feature vector outputs in these cases were combined using simple addition and fed as an input to a random forest classifier for further classification. The simple addition led to a lack of feature reduction techniques in this case which also caused non-significant features to be utilized for classification. Further, classification from direct OCT volume employing a 3D-CNN without extracting RNFL and GCC thickness reduced complexity and gave satisfactory performance, but didn't use textural details for classification. Also, classification from direct OCT volume using gNet3D with three convolutional and two FCN layers doesn't include internal textural details present within the slices due to classification on direct OCT. Further, using 3D-CNN with activation maps which could be

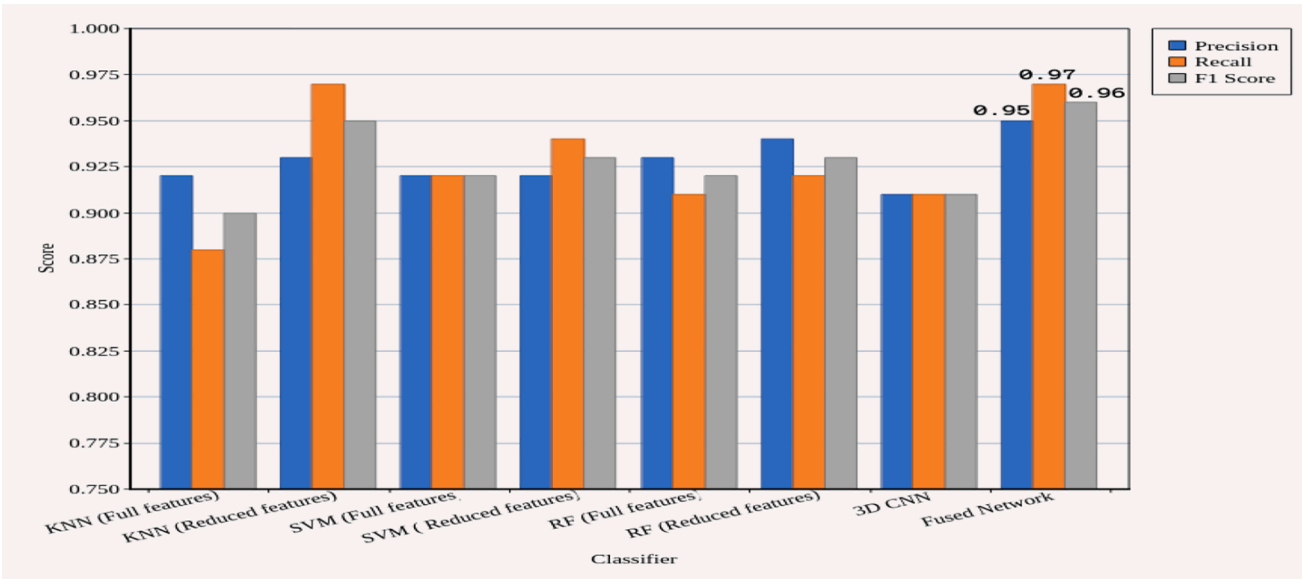


Fig. 9. Comparison of classifiers on reduced and full features.

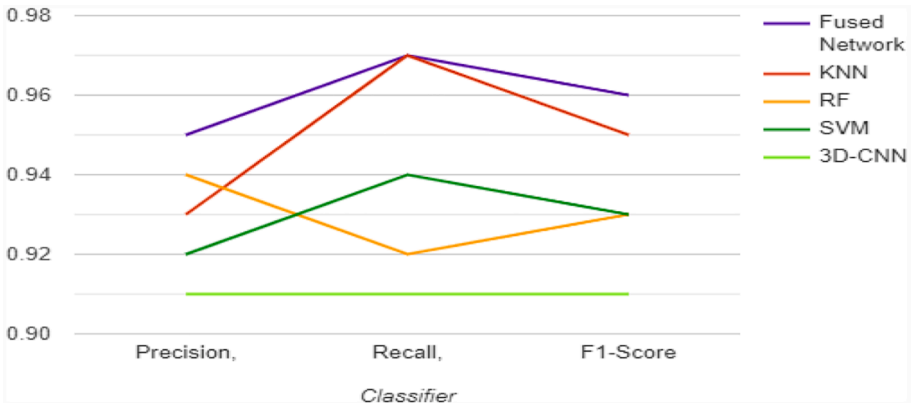


Fig. 10. Comparison of classifiers on Reduced Features.

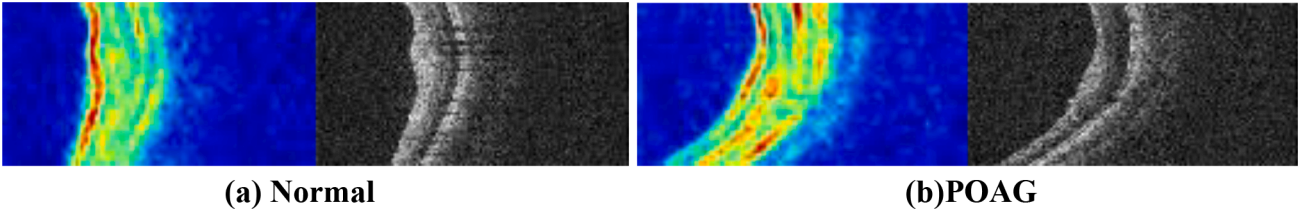


Fig. 11. Axial View.

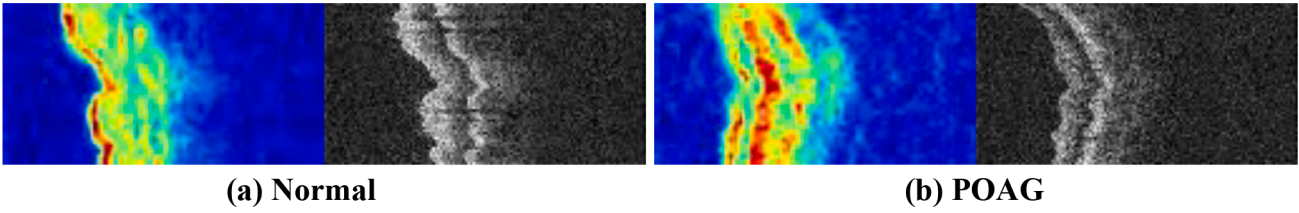


Fig. 12. Coronal View.

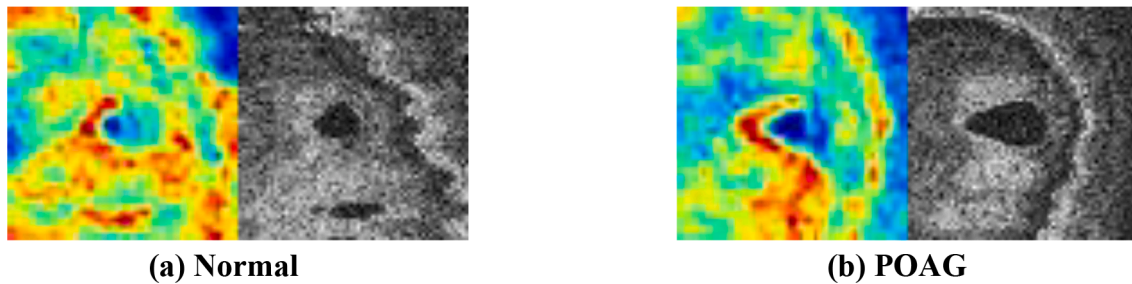


Fig. 13. Sagittal View.

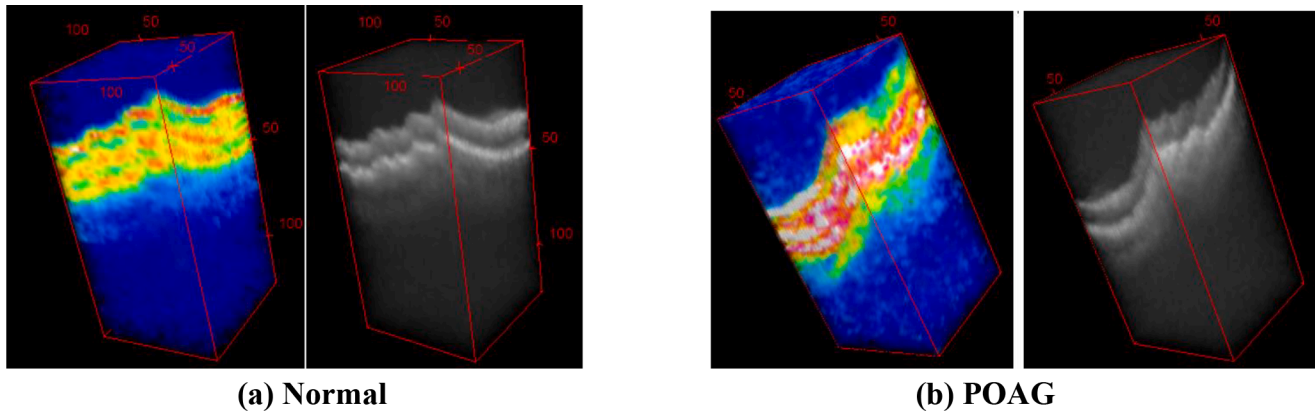


Fig. 14. Axial Voxel View.

utilized by clinicians for analyzing areas affected in images as input instead of direct OCT volume missed internal surface details to present within slices. Thus, the proposed model offers improved performance as compared to classification using both machine learning and deep learning. The high F1-Score in this case indicated effective performance of the proposed approach even against an uneven number of normal and abnormal samples in the dataset. Finally, the methodology used in this study differs from others, as it uses a fusion of machine learning approaches namely KNN, RF, SVM and 3D-CNN based deep learning. The KNN classifier used in this case offers robustness to noise, while RF offered effectiveness against missing data (if any) and SVM classifier was used due to effectiveness of binary classification (normal and glaucomatous in this case). The training of machine learning classifiers was performed using features such as Statistical, GLCM and GLRM. The huge amount of different features extracted in this case overcomes the drawback of missing internal surface details which were not used in the similar other approaches. Additionally, feature selection by Information Gain, Gain Ratio and Correlation techniques further provides the best set of features for classification which overcomes the drawback of non-significant features being used for classification. The use of 3DCNN thereafter performed direct classification on OCT volume. Finally, decision fusion of different classifiers with weighted fusion and major voting ensured effective classification with minimum error. Moreover, the use of Grad-CAM gave a visual representation, which could assist clinicians in validating output by noting the region of interest and further performing their own diagnosis within refined areas.

Thus, the contribution in the study could be summarized as the combination of deep learning and machine learning in the domain of CAD for glaucoma. To ensure the best performance of both techniques, 513 statistical features were extracted to be further reduced by feature selection techniques to ensure the best performance of machine learning models deployed (Section 2.3). Different parameters for each model were tested to eliminate overfitting and hence making the approach more robust for different datasets. For deep learning, augmentation

techniques were used to reduce overfitting and make the trained model more adaptive for OCT modality. The fusion was done using Major Voting and Weighted Decision Fusion to explore different combinations for fusing the approaches. After that the fusion was tested using metrics such as precision, recall and F1-Score. The AUROC curve is also studied to remove bias and variance in the fused framework approach (Fig. 8). To ensure that the fused framework was able to rectify incorrect predictions, the metric *dispute* was introduced, which measures the number of input samples for which dissimilar classes were predicted by the classifiers, hence the goal of the fused frame was set to reduce the number of disputes. Evaluation metrics were monitored based on the threshold probability and disputes were minimised to ensure optimum fusion of the above method (Fig. 7) (Section 2.7.3). To comprehensively understand the Deep Learning model, Grad-CAM was added based on the previous review which made the approach thorough (Fig. 13). The end result is a robust framework for classification of glaucoma using OCT volumes.

6. Conclusion

Glaucoma as a disorder of the retina has badly influenced the vision of people in a huge section across the world and is likely to influence nearly 110 million individuals by 2040. Diagnosis at an initial stage is a crucial facet for the prevention of further vision loss, because if not performed early, it might lead to enduring vision loss. Owing to the traditional diagnostic methods, which are primarily manual and time consuming for producing accurate results, often leads to delayed diagnosis and ultimately vision loss. Whereas, existing techniques employed till date provide less accuracy and success rate, which leads to a high number of affected people. Further, the performance as per state-of-the-art approaches is not satisfactory and hence is not acceptable clinically. Hence, an improved methodology has been proposed in this paper, wherein the Precision of 0.95, Recall of 0.97 and F1-Score of 0.96 was observed for the fused framework. The proposed fusion framework

shows improved performance of diagnosis, which is better than the existing studies and could be accepted by clinicians for prediction. Further, the proposed framework can be evaluated on extensive clinical images in local as well as real-time scenarios, thereby making it useful for initial screening of glaucoma by the experts.

CRedit authorship contribution statement

Mamta Juneja: Conceptualization, Project administration, Supervision, Writing – review & editing, Resources. **Janmejai Singh Minhas:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – review & editing, Software, Visualization. **Naveen Singla:** Conceptualization, Methodology, Formal analysis, Software, Investigation, Visualization, Writing – review & editing. **Sarthak Thakur:** Conceptualization, Writing – review & editing. **Niharika Thakur:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Data curation, Validation. **Prashant Jindal:** Project administration, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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